

Ten examples where you need Exercise 3

θ sits inside every term: no explicit estimator, no continuous-mapping shortcut

Why these examples need Exercise 3

In each example the estimating function has the form $\Psi_n(\theta) = \frac{1}{n} \sum_i \psi_\theta(Z_i)$ where θ is *inside* each term in a way that cannot be separated (e.g. Z_i^k , $\sigma(\beta X_i)$, $\tanh(Z_i - \theta)$, $\mathbf{1}_{(Z_i \leq \theta)}$). So Ψ_n is not a function of a few θ -free averages, $\hat{\theta}_n$ has no explicit form, and “LLN + continuous mapping” does not give consistency. Exercise 3 does: it only needs the LLN at two fixed points plus a shape property of Ψ_n .

Format. *Setup* (m , $\psi = \partial_\theta m$, $\partial_\theta \psi$, Ψ_n , Ψ) $\rightarrow$ *Step 1* Exercise 3(a) [(a1) continuous, exactly one zero; (a2) LLN at fixed θ ; (a3) sign change] or 3(b) [(b1) monotone; (b2) LLN; (b3) sign change; (b4) near-zero] $\rightarrow$ *Step 2* Theorem 4.2 C1–C4 $\rightarrow$ *Step 3* Σ/H_0^2 . Decreasing Ψ_n : verify for $-\Psi_n$, $-\Psi$.

	Model	Term with θ inside	Ex. 3	Variance
1	Weibull shape	$Z_i^k \log Z_i$	(a)	$k_0^2/(\pi^2/6 + (1 - \gamma)^2)$
2	Logistic location (MLE)	$\tanh((Z_i - \theta)/2)$	(a)	3
3	Claim-severity regression (Exp, log link)	$Z_i X_i e^{\beta X_i}$	(a)	$1/\mathbb{E} X^2$
4	Poisson regression, 1-D	$e^{\beta X_i} X_i$	(b)	$1/\mathbb{E}[e^{\beta_0 X} X^2]$
5	Logistic regression, 1-D	$\sigma(\beta X_i) X_i$	(b)	$1/\mathbb{E}[\sigma_0(1 - \sigma_0) X^2]$
6	Mixture weight	$\frac{f_1 - f_0}{\pi f_1 + (1 - \pi) f_0}(Z_i)$	(b)	$1/\int \frac{(f_1 - f_0)^2}{g_{\pi_0}}$
7	Log-cosh location (non-MLE)	$\tanh(Z_i - \theta)$	(a)	$\mathbb{E} \tanh^2 / (\mathbb{E} \operatorname{sech}^2)^2$
8	Huber location (non-smooth)	$\psi_k(Z_i - \theta)$	(b)	$\mathbb{E} \psi_k^2 / \mathbb{P}(Z - \theta_0 < k)^2$
9	τ -quantile (non-smooth)	$\mathbf{1}_{(Z_i \leq \theta)}$	(b)	$\tau(1 - \tau)/f(\theta_0)^2$
10	Laplace location MLE (non-smooth)	$\operatorname{sign}(Z_i - \theta)$	(b)	$1 = I^{-1}$

Example 1: Weibull shape

Z_i iid with density $f(z | k) = kz^{k-1}e^{-z^k}$, $z > 0$, $k \in (0, \infty)$. Find the asymptotic distribution of the MLE.

Setup. $\ell = \log k + (k-1) \log z - z^k$, $\psi_k = \frac{1}{k} + \log z - z^k \log z$, $\partial_k \psi_k = -\frac{1}{k^2} - z^k (\log z)^2 < 0$.
 Useful fact: $W = Z^{k_0} \sim \text{Exp}(1)$ and $\log Z = \log W/k_0$; $\mathbb{E} \log W = -\gamma$, $\mathbb{E}[W \log W] = 1 - \gamma$, $\mathbb{E}[W(\log W)^2] = \frac{\pi^2}{6} - 1 + (1 - \gamma)^2$.

Why Exercise 3: the likelihood equation $\frac{1}{k} + \overline{\log Z} - \frac{1}{n} \sum Z_i^k \log Z_i = 0$ has k in the exponent inside the sum.

Step 1: consistency via Exercise 3(a). Verify for $-\Psi_n$.

- (a1) $\partial \psi < 0$: Ψ_n is continuous and strictly decreasing. As $k \downarrow 0$, $\frac{1}{k} \rightarrow \infty$ while the other terms stay bounded, so $\Psi_n \rightarrow +\infty$; as $k \rightarrow \infty$, $Z_i^k \log Z_i \rightarrow \infty$ for any $Z_i > 1$, so $\Psi_n \rightarrow -\infty$ once some $Z_i > 1$ (probability $\rightarrow 1$). Exactly one zero.
- (a2) $\mathbb{E}|\psi_k(Z)| \leq \frac{1}{k} + \mathbb{E}|\log Z| + \mathbb{E}[Z^k |\log Z|] < \infty$ (all moments of W , $\log W$ exist). LLN.
- (a3) $\Psi(k_0) = 0$ (C3 below) and $\Psi' = \mathbb{E} \partial \psi < 0$: $\Psi(k_0 - \varepsilon) > 0 > \Psi(k_0 + \varepsilon)$.

By Exercise 3(a), $\hat{k}_n \xrightarrow{\mathbb{P}} k_0$.

Step 2: conditions of Theorem 4.2.

C2 C^∞ in k ; support $(0, \infty)$; Θ open.

C3 $\mathbb{E} \psi_{k_0} = \frac{1}{k_0} + \frac{\mathbb{E} \log W - \mathbb{E}[W \log W]}{k_0} = \frac{1 - \gamma - (1 - \gamma)}{k_0} = 0$; finite variance.

C4 On $[k_0/2, 2k_0]$: $z^k \leq z^{k_0/2} + z^{2k_0}$, so $|\partial \psi| \leq \frac{4}{k_0^2} + (z^{k_0/2} + z^{2k_0})(\log z)^2 =: d(z)$, integrable.
 $H_0 = -I$.

Step 3: compute and conclude. $I = -\mathbb{E} \partial \psi = \frac{1 + \mathbb{E}[W(\log W)^2]}{k_0^2} = \frac{\pi^2/6 + (1 - \gamma)^2}{k_0^2} \approx \frac{1.824}{k_0^2}$.

$$\sqrt{n}(\hat{k}_n - k_0) \xrightarrow{d} N(0, k_0^2/(\pi^2/6 + (1 - \gamma)^2)) \approx N(0, 0.548 k_0^2).$$

Example 2: Logistic location model

Z_i iid with logistic density $f(z | \theta) = \frac{e^{-(z-\theta)}}{(1 + e^{-(z-\theta)})^2}$, $z \in \mathbb{R}$. Find the asymptotic distribution of the MLE. ($\sigma(u) = 1/(1 + e^{-u})$.)

Setup. With $u = z - \theta$: $\ell = -u - 2 \log(1 + e^{-u})$, $\frac{\partial \ell}{\partial u} = -1 + 2(1 - \sigma(u)) = 1 - 2\sigma(u)$, so

$$\psi_\theta(z) = -\frac{\partial \ell}{\partial u} = 2\sigma(z - \theta) - 1 = \tanh\left(\frac{z - \theta}{2}\right), \quad \partial_\theta \psi_\theta = -2\sigma(z - \theta)(1 - \sigma(z - \theta)) \in [-\frac{1}{2}, 0).$$

Useful fact: if $U = Z - \theta_0$ is standard logistic, then $V = \sigma(U) \sim \text{Unif}(0, 1)$, since $\mathbb{P}(\sigma(U) \leq s) = \sigma(\sigma^{-1}(s)) = s$.

Why Exercise 3: the likelihood equation $\sum_i \tanh((Z_i - \theta)/2) = 0$ has θ inside a nonlinear function of each Z_i .

Step 1: consistency via Exercise 3(a). Verify for $-\Psi_n$.

(a1) Ψ_n is continuous, strictly decreasing ($\partial\psi < 0$), from $+1$ ($\theta \rightarrow -\infty$) to -1 ($\theta \rightarrow \infty$): exactly one zero.

(a2) $|\psi| \leq 1$: LLN.

(a3) $\Psi(\theta_0) = \mathbb{E}[2V - 1] = 0$; $\Psi'(\theta) = -\mathbb{E}[2\sigma(1 - \sigma)(Z - \theta)] < 0$: sign change.

By Exercise 3(a), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Step 2: conditions of Theorem 4.2.

C2 C^∞ ; support $\mathbb{R}$; $\Theta = \mathbb{R}$.

C3 Mean 0; $\Sigma = \mathbb{E}(2V - 1)^2 = \text{Var}(2V) = 4 \cdot \frac{1}{12} = \frac{1}{3}$.

C4 $|\partial\psi| \leq \frac{1}{2}$: constant d . $H_0 = -2\mathbb{E}[V(1 - V)] = -2(\frac{1}{2} - \frac{1}{3}) = -\frac{1}{3}$.

Step 3: compute and conclude. $\Sigma = -H_0 = I = \frac{1}{3}$, so the variance is 3.

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, 3)$. Compare: sample mean $\text{Var}(U) = \pi^2/3 \approx 3.29$; median $1/(4f(0)^2) = 4$.

Example 3: Claim-severity regression (exponential, log link)

(X_i, Z_i) iid, $|X| \leq K$, $\mathbb{P}(X > 0) > 0$, $\mathbb{P}(X < 0) > 0$, and $Z \mid X \sim \text{Exp}$ with rate $e^{\beta_0 X}$ (mean claim size $e^{-\beta_0 X}$). Find the asymptotic distribution of the MLE of β .

Setup. $\ell = \beta X - Ze^{\beta X}$, $\psi_\beta = X(1 - Ze^{\beta X})$, $\partial_\beta \psi_\beta = -ZX^2 e^{\beta X} \leq 0$.
 Useful fact: given X , $Ze^{\beta_0 X} \sim \text{Exp}(1)$, so $\mathbb{E}[Z \mid X] = e^{-\beta_0 X}$, $\text{Var}(Ze^{\beta_0 X} \mid X) = 1$.
 $\Psi(\beta) = \mathbb{E}[X(1 - e^{\beta X} \mathbb{E}[Z \mid X])] = \mathbb{E}[X(1 - e^{(\beta - \beta_0)X})]$.

Why Exercise 3: $\sum_i X_i(1 - Z_i e^{\beta X_i}) = 0$ has β inside $e^{\beta X_i}$ for each i .

Step 1: consistency via Exercise 3(a). Verify for $-\Psi_n$.

- (a1) Ψ_n is continuous and strictly decreasing once some $X_i \neq 0$. As $\beta \rightarrow \infty$, terms with $X_i > 0$ give $-Z_i X_i e^{\beta X_i} \rightarrow -\infty$; as $\beta \rightarrow -\infty$, terms with $X_i < 0$ give $+Z_i |X_i| e^{\beta X_i} \rightarrow +\infty$. With probability $\rightarrow 1$ both signs of X occur: exactly one zero.
- (a2) $|\psi_\beta| \leq K(1 + Ze^{|\beta|K})$, and $\mathbb{E}Z \leq e^{|\beta_0|K}$: LLN.
- (a3) $\Psi(\beta_0) = 0$; $\Psi'(\beta) = -\mathbb{E}[X^2 e^{(\beta - \beta_0)X}] < 0$: sign change.

By Exercise 3(a), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

Step 2: conditions of Theorem 4.2.

C2 Smooth in β ; support $(0, \infty)$; $\Theta = \mathbb{R}$.

C3 $\mathbb{E}\psi_{\beta_0} = \mathbb{E}[X(1 - \mathbb{E}[Ze^{\beta_0 X} \mid X])] = 0$; $\Sigma = \mathbb{E}[X^2 \text{Var}(Ze^{\beta_0 X} \mid X)] = \mathbb{E}X^2$.

C4 On $[\beta_0 - 1, \beta_0 + 1]$: $|\partial\psi| \leq XK^2 e^{(|\beta_0|+1)K} =: d$, $\mathbb{E}d < \infty$. $H_0 = -\mathbb{E}[X^2 e^{\beta_0 X} \mathbb{E}[Z \mid X]] = -\mathbb{E}X^2$.

Step 3: compute and conclude. $\Sigma = -H_0 = \mathbb{E}X^2$.

$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}X^2)$ (free of β_0).

Example 4: Poisson regression with one covariate

(X_i, Y_i) iid, $|X| \leq K$, $\mathbb{P}(X \neq 0) > 0$, $Y \mid X \sim \text{Poisson}(e^{\beta_0 X})$. Find the asymptotic distribution of the MLE.

Setup. $\ell = Y\beta X - e^{\beta X} - \log Y!$, $\psi_\beta = (Y - e^{\beta X})X$, $\partial_\beta \psi_\beta = -e^{\beta X} X^2 \leq 0$.
 $\Psi(\beta) = \mathbb{E}[(e^{\beta_0 X} - e^{\beta X})X]$ (tower property).

Why Exercise 3: $\sum_i (Y_i - e^{\beta X_i})X_i = 0$ has β inside $e^{\beta X_i}$.

Step 1: consistency via Exercise 3(b). Verify for $-\Psi_n$.

- (b1) $\partial\psi \leq 0$: Ψ_n nonincreasing.
- (b2) $|\psi| \leq K(Y + e^{|\beta|K})$, $\mathbb{E}Y \leq e^{|\beta_0|K}$: LLN.
- (b3) $\Psi(\beta_0) = 0$, $\Psi' = -\mathbb{E}[e^{\beta X} X^2] < 0$: sign change.
- (b4) Ψ_n is continuous; with probability $\rightarrow 1$, $\Psi_n(\beta_0 - \varepsilon) > 0 > \Psi_n(\beta_0 + \varepsilon)$ (by (b2)–(b3)), so a root exists; $\Psi_n(\hat{\beta}_n) = 0$.

By Exercise 3(b), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

Step 2: conditions of Theorem 4.2.

C2 Smooth; support $\{0, 1, \dots\}$; $\Theta = \mathbb{R}$.

C3 $\mathbb{E}\psi_{\beta_0} = 0$; $\Sigma = \mathbb{E}[X^2 \text{Var}(Y \mid X)] = \mathbb{E}[e^{\beta_0 X} X^2]$.

C4 $|\partial\psi| \leq K^2 e^{(|\beta_0|+1)K}$ on $[\beta_0 - 1, \beta_0 + 1]$. $H_0 = -\mathbb{E}[e^{\beta_0 X} X^2]$.

$$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}[e^{\beta_0 X} X^2]).$$

Example 5: Logistic regression with one covariate

(X_i, Y_i) iid, $\mathbb{E}X^2 < \infty$, $\mathbb{P}(X \neq 0) > 0$, $Y \mid X \sim \text{Bernoulli}(\sigma(\beta_0 X))$. Find the asymptotic distribution of the MLE.

Setup. $\ell = Y\beta X - \log(1 + e^{\beta X})$, $\psi_\beta = (Y - \sigma(\beta X))X$, $\partial_\beta \psi_\beta = -\sigma(1 - \sigma)(\beta X) X^2 \leq 0$.
 $\Psi(\beta) = \mathbb{E}[(\sigma(\beta_0 X) - \sigma(\beta X))X]$; write $\sigma_0 = \sigma(\beta_0 X)$.

Why Exercise 3: $\sum_i (Y_i - \sigma(\beta X_i))X_i = 0$ has β inside $\sigma(\beta X_i)$; moreover the MLE need not exist (perfect separation), so we use the near-zero version 3(b).

Step 1: consistency via Exercise 3(b). Verify for $-\Psi_n$.

- (b1) $\partial\psi \leq 0$: nonincreasing.
- (b2) $|\psi| \leq |X|$, integrable: LLN.
- (b3) $\Psi(\beta_0) = 0$, $\Psi' = -\mathbb{E}[\sigma(1 - \sigma)(\beta X)X^2] < 0$.
- (b4) Continuous; sign change at the posts with probability $\rightarrow 1$, so a root exists; $\Psi_n(\hat{\beta}_n) = 0$.

By Exercise 3(b), $\hat{\beta}_n \xrightarrow{\mathbb{P}} \beta_0$.

Step 2: conditions of Theorem 4.2.

C2 Smooth; $Y \in \{0, 1\}$; $\Theta = \mathbb{R}$.

C3 $\mathbb{E}\psi_{\beta_0} = \mathbb{E}[X(\mathbb{E}[Y \mid X] - \sigma_0)] = 0$; $\Sigma = \mathbb{E}[X^2\sigma_0(1 - \sigma_0)]$.

C4 $|\partial\psi| \leq \frac{1}{4}X^2$. $H_0 = -\mathbb{E}[\sigma_0(1 - \sigma_0)X^2]$.

$\sqrt{n}(\hat{\beta}_n - \beta_0) \xrightarrow{d} N(0, 1/\mathbb{E}[\sigma_0(1 - \sigma_0)X^2])$. E.g. $X = \pm 1$ equally likely, $\beta_0 = 0$: variance 4.

Example 6: Weight of a two-component mixture

Claims come from a “normal” density f_0 or a “large-claim” density f_1 , both known: Z_i iid with density $g_\pi = \pi f_1 + (1 - \pi)f_0$, $\pi_0 \in (0, 1)$, and $f_1 \neq f_0$ on a set of positive measure. Find the asymptotic distribution of the MLE of π .

Setup. $\ell = \log g_\pi(z)$, $\psi_\pi = \frac{f_1(z) - f_0(z)}{g_\pi(z)}$, $\partial_\pi \psi_\pi = -\frac{(f_1 - f_0)^2}{g_\pi^2} = -\psi_\pi^2 \leq 0$.

Bound: $g_\pi \geq \min(\pi, 1 - \pi)(f_0 + f_1) \geq \min(\pi, 1 - \pi)|f_1 - f_0|$, so $|\psi_\pi| \leq 1/\min(\pi, 1 - \pi)$.

Why Exercise 3: $\sum_i \frac{f_1(Z_i) - f_0(Z_i)}{\pi f_1(Z_i) + (1 - \pi)f_0(Z_i)} = 0$ has π in the denominator of each term; and the root need not exist in $(0, 1)$ for every sample, so we use 3(b).

Step 1: consistency via Exercise 3(b). Verify for $-\Psi_n$ on $(0, 1)$.

- (b1) $\partial\psi = -\psi^2 \leq 0$: nonincreasing.
- (b2) $|\psi_\pi| \leq 1/\min(\pi, 1 - \pi)$: bounded, LLN.
- (b3) $\Psi(\pi_0) = \int \frac{f_1 - f_0}{g_{\pi_0}} g_{\pi_0} dz = \int f_1 - \int f_0 = 1 - 1 = 0$; $\Psi'(\pi) = -\mathbb{E}_{\pi_0}[\psi_\pi^2] < 0$ (as $f_1 \neq f_0$; differentiation under $\mathbb{E}$ is fine because ψ^2 is bounded on compact subintervals). Sign change.
- (b4) Ψ_n continuous; sign change at the posts with probability $\rightarrow 1$, so a root exists in the window; $\Psi_n(\hat{\pi}_n) = 0$.

By Exercise 3(b), $\hat{\pi}_n \xrightarrow{\mathbb{P}} \pi_0$.

Step 2: conditions of Theorem 4.2.

C2 Smooth in $\pi \in (0, 1)$; support = $\text{supp} f_0 \cup \text{supp} f_1$ for every $\pi \in (0, 1)$; π_0 interior.

C3 Mean 0 by (b3); $\Sigma = \mathbb{E} \psi_{\pi_0}^2 = \int \frac{(f_1 - f_0)^2}{g_{\pi_0}} dz$, finite since ψ is bounded.

C4 On $[\pi_0 - \delta, \pi_0 + \delta] \subset (0, 1)$: $|\partial \psi| = \psi^2 \leq 1 / \min(\pi_0 - \delta, 1 - \pi_0 - \delta)^2$, a constant. $H_0 = -\Sigma$.

$\sqrt{n}(\hat{\pi}_n - \pi_0) \xrightarrow{d} N\left(0, \left[\int \frac{(f_1 - f_0)^2}{\pi_0 f_1 + (1 - \pi_0) f_0}\right]^{-1}\right)$. Example $f_0 = \text{Exp}(1)$, $f_1 = \text{Exp}(0.2)$, $\pi_0 = 0.2$: $I \approx 2.41$, variance ≈ 0.41 (numerical integration).

Example 7: Log-cosh location (not an MLE)

Z_i iid with a density symmetric about θ_0 . $\hat{\theta}_n$ maximises $-\frac{1}{n} \sum_i \log \cosh(Z_i - \theta)$. Find its asymptotic distribution.

Setup. $m = -\log \cosh(z - \theta)$, $\psi_\theta = \tanh(z - \theta)$, $\partial_\theta \psi_\theta = -\operatorname{sech}^2(z - \theta) \in [-1, 0)$.

Why Exercise 3: $\sum_i \tanh(Z_i - \theta) = 0$ cannot be rewritten through finitely many θ -free averages.

Step 1: consistency via Exercise 3(a). Verify for $-\Psi_n$.

- (a1) Continuous, strictly decreasing from +1 to -1: exactly one zero.
- (a2) $|\tanh| \leq 1$: LLN.
- (a3) Symmetry and oddness: $\Psi(\theta_0) = 0$; $\Psi' = -\mathbb{E} \operatorname{sech}^2(Z - \theta) < 0$.

By Exercise 3(a), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Step 2: conditions of Theorem 4.2.

C2 C^∞ ; $\Theta = \mathbb{R}$. C3: mean 0; $\Sigma = \mathbb{E} \tanh^2(Z - \theta_0)$. C4: $|\partial \psi| \leq 1$; $H_0 = -\mathbb{E} \operatorname{sech}^2(Z - \theta_0)$.

Step 3: compute and conclude. Not an MLE: keep Σ/H_0^2 .

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \mathbb{E} \tanh^2(Z - \theta_0)/(\mathbb{E} \operatorname{sech}^2(Z - \theta_0))^2)$. For $N(\theta_0, 1)$: $0.394/0.606^2 \approx 1.075$.

Example 8: Huber's location estimator (non-smooth)

Z_i iid with continuous density f symmetric about θ_0 , $f > 0$ near θ_0 . Fix $k > 0$, $\psi_k(u) = \max(-k, \min(k, u))$. $\hat{\theta}_n$ solves $\sum_i \psi_k(Z_i - \theta) = 0$. Find its asymptotic distribution.

Setup. $\psi_\theta(z) = \psi_k(z - \theta)$; it is continuous and Lipschitz in θ with constant 1, but its θ -derivative $-\mathbf{1}_{(|z-\theta|<k)}$ jumps at $\theta = z \pm k$.

Why Exercise 3: θ is inside the clipping of every term; the zero set may be an interval (if all residuals exceed k), so we use 3(b), which needs no uniqueness.

Step 1: consistency via Exercise 3(b). Verify for $-\Psi_n$.

- (b1) ψ_k nondecreasing and $z - \theta$ decreasing in θ : Ψ_n nonincreasing.
- (b2) $|\psi_k| \leq k$: LLN.
- (b3) Symmetry, ψ_k odd: $\Psi(\theta_0) = 0$; $\Psi'(\theta) = -\mathbb{P}(|Z - \theta| < k) < 0$ near θ_0 .
- (b4) Ψ_n continuous with limits $\pm k$: a zero exists, $\Psi_n(\hat{\theta}_n) = 0$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Step 2: limit via the Exercise 7 route (C2/C4 fail: ψ is not differentiable).

- *Population slope:* $H_0 = \Psi'(\theta_0) = -\mathbb{P}(|Z - \theta_0| < k)$, continuous in θ since F is continuous.
- *CLT at θ_0 :* $\Sigma = \mathbb{E} \psi_k^2(Z - \theta_0) \leq k^2$.
- *Donsker:* $|\psi_{\theta_1} - \psi_{\theta_2}| \leq |\theta_1 - \theta_2|$, so on $[\theta_0 - C, \theta_0 + C]$ brackets $[\psi_{\theta_j} - \frac{\varepsilon}{2}, \psi_{\theta_j} + \frac{\varepsilon}{2}]$ with spacing $\varepsilon/2$ have L^2 -size ε ; $N_{[]} \approx 4C/\varepsilon$, $J_{[]} < \infty$; Theorem 1.19. Hence $\mathbb{G}_n \psi_{\hat{\theta}_n} - \mathbb{G}_n \psi_{\theta_0} = o_{\mathbb{P}}(1)$.
- *Expansion:* $0 = \sqrt{n} \Psi_n(\hat{\theta}_n) = \mathbb{G}_n \psi_{\theta_0} + o_{\mathbb{P}}(1) + \Psi'(\bar{\theta}_n) \sqrt{n}(\hat{\theta}_n - \theta_0)$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \mathbb{E}\psi_k(Z - \theta_0)^2 / \mathbb{P}(|Z - \theta_0| < k)^2). \text{ For } N(\theta_0, 1), k = 1: 0.516/0.683^2 \approx 1.107.$$

Example 9: The sample τ -quantile (non-smooth)

$\tau \in (0, 1)$ fixed; Z_i iid with cdf F , density f continuous and > 0 near $\theta_0 = F^{-1}(\tau)$. $\hat{\theta}_n$ minimises $\sum_i \rho_\tau(Z_i - \theta)$, $\rho_\tau(u) = u(\tau - \mathbf{1}_{(u < 0)})$. Find its asymptotic distribution.

Setup. $\psi_\theta(z) = \mathbf{1}_{(z \leq \theta)} - \tau$ (from $\partial_\theta \rho_\tau(z - \theta) = -\tau$ for $\theta < z$ and $1 - \tau$ for $\theta > z$); $\Psi_n = F_n - \tau$, $\Psi = F - \tau$.

Why Exercise 3: $\Psi_n(\theta) = F_n(\theta) - \tau$ is a different average for every θ , and a staircase: it has no exact zero in general, so 3(a) fails and 3(b) is needed.

Step 1: consistency via Exercise 3(b).

- (b1) F_n nondecreasing. (b2) Indicators bounded: LLN.
(b3) $f > 0$ near θ_0 : $F(\theta_0 - \varepsilon) < \tau < F(\theta_0 + \varepsilon)$.
(b4) Left/right derivatives of the convex criterion: $F_n(\hat{\theta}_n -) \leq \tau \leq F_n(\hat{\theta}_n)$; no ties, jump $\leq \frac{1}{n}$:
 $0 \leq \Psi_n(\hat{\theta}_n) \leq \frac{1}{n}$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Step 2: limit via the Exercise 7 route (C2/C4 fail: ψ is not differentiable).

- $H_0 = \Psi'(\theta_0) = f(\theta_0)$; $\Sigma = \text{Var}(\mathbf{1}_{(Z \leq \theta_0)}) = \tau(1 - \tau)$.
- Donsker: grid with F -increments $\leq \varepsilon^2$, brackets $[\psi_{t_{j-1}}, \psi_{t_j}]$, $N_{[]} \leq 1/\varepsilon^2 + 1$, Theorem 1.19.
- Expansion: $O(n^{-1/2}) = \mathbb{G}_n \psi_{\theta_0} + o_{\mathbb{P}}(1) + f(\bar{\theta}_n) \sqrt{n}(\hat{\theta}_n - \theta_0)$.

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \tau(1 - \tau)/f(\theta_0)^2).$$

Example 10: The Laplace location MLE (an MLE with a kink)

Z_i iid with density $f(z | \theta) = \frac{1}{2}e^{-|z - \theta|}$. Find the asymptotic distribution of the MLE.

Setup. $\ell = -\log 2 - |z - \theta|$, $\psi_\theta(z) = \text{sign}(z - \theta)$ (for $z \neq \theta$). Maximising $\ell_n = \text{minimising } \sum |Z_i - \theta|$: the MLE is the sample median.

Why Exercise 3: $\sum_i \text{sign}(Z_i - \theta)$ is a staircase in θ with no exact zero for odd n ; and ℓ has a kink at $\theta = z$, so Theorem 4.3 cannot be applied directly.

Step 1: consistency via Exercise 3(b). $\Psi_n(\theta) = \frac{1}{n} \sum \text{sign}(Z_i - \theta)$ is nonincreasing; verify for $-\Psi_n(\theta) = \frac{1}{n} \sum \text{sign}(\theta - Z_i)$.

- (b1) Each $\text{sign}(\theta - Z_i)$ is nondecreasing in θ .
(b2) Bounded: LLN; $-\Psi(\theta) = 2F(\theta) - 1$.
(b3) The Laplace cdf is strictly increasing, $F(\theta_0) = \frac{1}{2}$: sign change.
(b4) No ties; counting below/above the median: $|\Psi_n(\hat{\theta}_n)| \leq \frac{1}{n}$.

By Exercise 3(b), $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta_0$.

Step 2: limit via the Exercise 7 route (C2/C4 fail: ψ is not differentiable).

- $\Psi(\theta) = \mathbb{E} \text{sign}(Z - \theta) = 1 - 2F(\theta)$, so $H_0 = \Psi'(\theta_0) = -2f(\theta_0 | \theta_0) = -2 \cdot \frac{1}{2} = -1$.

- $\Sigma = \mathbb{E} \text{sign}^2 = 1$; note this equals the Fisher information $I = \mathbb{E}[\psi^2] = 1$.
- Donsker for $\{\text{sign}(\cdot - \theta)\} = \{1 - 2 \cdot \mathbf{1}_{(\cdot \leq \theta)}\}$: the indicator brackets of Example 9.

$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Sigma/H_0^2) = N(0, 1) = N(0, I^{-1})$. The MLE still attains I^{-1} , but the proof needs the non-smooth route.

Summary: what made each one work

Ex.	Shape property used	Near-zero / unique zero	Normality step
1 Weibull	strictly decreasing ($\ell'' < 0$)	limits $\pm\infty$: one zero	Thm 4.3, C4 by worst endpoint
2 Logistic location	strictly decreasing, bounded	limits ± 1 : one zero	Thm 4.3, $ \partial\psi \leq \frac{1}{2}$
3 Severity GLM	strictly decreasing	limits $\pm\infty$ when X takes both signs	Thm 4.3, $d = ZK^2e$
4 Poisson GLM	nonincreasing	root exists w.p. $\rightarrow 1$ (posts + IVT)	Thm 4.3, constant d
5 Logistic GLM	nonincreasing	root exists w.p. $\rightarrow 1$ (separation possible)	Thm 4.3, $d = \frac{1}{4}X^2$
6 Mixture weight	$\partial\psi = -\psi^2 \leq 0$	root exists w.p. $\rightarrow 1$	Thm 4.3, constant d
7 Log-cosh	strictly decreasing, bounded	limits ± 1 : one zero	Thm 4.2, full sandwich
8 Huber	nonincreasing, Lipschitz	exact zero, maybe not unique	Ex. 7 route, Lipschitz brackets
9 Quantile	staircase, nondecreasing	jump $\leq 1/n$	Ex. 7 route, indicator brackets
10 Laplace MLE	staircase, nonincreasing	jump $\leq 1/n$	Ex. 7 route; answer $I^{-1} = 1$